

Mainak Debnath

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Summary

Backend-focused Software Engineer with 4 years of experience building and scaling product features using **ASP.NET Core**, **Angular**, and **SQL**. Experienced in backend integrations, asynchronous processing, API optimization, concurrent workflows, and delivering reliable production systems for enterprise applications.

Technical Skills

Languages: C#, SQL, TypeScript, Python

Backend: ASP.NET Core, REST APIs, SQL Server, Django, Redis, Message Queues, xUnit, Moq

Frontend: Angular, HTML/CSS

AI / Data: LangChain, Vector Databases, Agentic AI, RAG

Engineering: Git, GitHub Actions, CI/CD, Concurrency, Multithreading, Performance Optimization

Experience

MRI Software [🔗](#)

Bengaluru, India

Software Engineer 2

Feb 2025 – Present

- Owned end-to-end integration of **Blue Moon** into the Residential Management platform, designing and implementing backend workflows, production rollout, and operational stabilization.
- Re-architected bulk document processing into an **event-driven asynchronous workflow** using message-based execution, reducing end-to-end latency from **30s to 5s (83%)** while improving throughput under load.
- Improved processing throughput through controlled parallel execution using **multithreading**, **semaphore-based concurrency control**, and asynchronous execution while maintaining system stability.
- Led integration of **SecureSign** into the Affordable Housing platform, enabling digital document workflows and reducing document turnaround time by **70%**.

Software Engineer 1

Aug 2022 – Jan 2025

- Designed and delivered full-stack product modules using **Angular**, **ASP.NET Core**, and **SQL**, building UI workflows, REST APIs, and database changes for enterprise housing operations.
- Optimized high-volume Angular screens using **lazy loading**, pagination, and improved data-fetching strategies, reducing initial load time by **30%**.
- Improved backend API performance by refactoring service-layer logic and database access patterns, reducing API latency by **15%** and increasing request throughput.
- Designed and optimized **SQL queries** for CRUD-heavy modules, improving query performance and transactional reliability across business workflows.
- Implemented automated **unit and integration testing** across frontend and backend changes, collaborating closely with QA and product teams to accelerate sprint velocity and ensure reliable production delivery.

Projects

ReviewPal [🔗](#) | Python, LangChain, Gemini AI, ChromaDB, GitHub Actions, GitHub REST APIs

May 2025

- Built an **AI-powered GitHub pull request reviewer** that generates and posts **standards-aware inline feedback** using repository code context and project-specific coding rules.
- Engineered a **CI-triggered RAG pipeline** with **ChromaDB**, combining **semantic retrieval + lexical search**, same-file context prioritization, and **content-hash-based incremental indexing**.
- Improved reviewer quality and runtime through **parallel diff processing**, **second-pass verification**, **confidence-based ranking**, and validation guardrails for added-line accuracy and duplicate suppression.

IntelliBlogger [🔗](#) | Python, DRF, Angular, Redis, Celery, SQLite, Gemini API

Jan 2024 - March 2024

- Built a full-stack AI content platform using **Angular** and **Django REST Framework** to convert YouTube videos into editable long-form blog posts, with a **job-based generation pipeline** featuring persisted job state, retry handling, and resumable status tracking for long-running AI workloads.
- Implemented JWT authentication, endpoint-level **rate limiting**, and **Redis-backed caching**, reducing repeated generation latency from **25s to under 5s**, with **SSE-based** job status updates and **Celery-backed** async processing for production readiness.

Education

St. Thomas' College of Engineering and Technology

Kolkata, India

B.Tech in Computer Science and Engineering GPA : 9.08/10

Aug. 2018 – July 2022

Achievements

- Secured **2nd place** in the annual company-wide hackathon for building **ReviewPal**, an AI-powered automated code review agent integrated with GitHub PR workflows.
- Merged contributions into **NumPy** [🔗](#), **freeCodeCamp** [🔗](#), and **sButtons** [🔗](#), improving developer tooling and community-driven projects.